

National Textile University, Faisalabad



Department of Computer Science

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Class:	BSCS-A
Registration No:	23-NTU-CS-1047
Course Name:	Embedded Iot Systems
Submitted To:	Sir Nasir Mahmood
Date of Submission:	26 th October,2025

Assignment # 01

Questions 3:

Task # 02:

// Name: Maryam Shamshad

// Reg No: 23-NTU-CS-1047

// Smart Button Control with OLED Feedback

```
#include <Arduino.h>
```

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>
```

```
// ===== OLED SETUP =====
```

```
#define SCREEN_WIDTH 128
```

```
#define SCREEN_HEIGHT 64
```

```
#define OLED_ADDR 0x3C
```

```
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
```

```
// ===== PIN CONFIGURATION =====
```

```
#define LED_PIN 18 // LED output pin
```

```
#define BUZZER_PIN 27 // Buzzer output pin
```

```
#define BTN_PIN 33 // Push button input pin (with internal pull-up)
```

```
// ===== TIMING VARIABLES =====
```

```
unsigned long pressStart = 0; // Stores the time when button is pressed
```

```
bool buttonPressed = false; // Tracks if the button is currently pressed
```

```
bool ledState = false; // Current state of LED (ON/OFF)
```

```
const unsigned long LONG_PRESS_TIME = 1500; // Long press threshold (1.5 seconds)
```

```
// ===== OLED DISPLAY FUNCTION =====
```

```
// Displays two lines of text on the OLED screen
```

```
void updateOLED(const String &line1, const String &line2 = "") {
```

```
    display.clearDisplay();
```

```
    display.setTextSize(1);
```

```
    display.setTextColor(SSD1306_WHITE);
```

```
    display.setCursor(0, 10);
```

```
    display.println(line1);
```

```
    if (line2 != "") {
```

```
        display.setCursor(0, 30);
```

```
        display.println(line2);
```

```
    }
```

```
    display.display();
```

```
}

// ===== SHORT PRESS FUNCTION =====
// Toggles LED ON/OFF when button is briefly pressed
void handleShortPress() {
  ledState = !ledState; // Toggle LED state
  digitalWrite(LED_PIN, ledState);
  Serial.println("Short Press → Toggle LED");
  updateOLED("Short Press", ledState ? "LED: ON" : "LED: OFF");
}

// ===== LONG PRESS FUNCTION =====
// Activates buzzer for a short tone when button is held
void handleLongPress() {
  Serial.println("Long Press → Play Buzzer");
  updateOLED("Long Press", "Buzzer Tone...");

  tone(BUZZER_PIN, 1000, 400); // Generate 1 kHz tone for 400 ms
  delay(400);
  noTone(BUZZER_PIN);          // Stop buzzer tone

  updateOLED("Long Press", "Done");
}

// ===== SETUP FUNCTION =====
void setup() {
  Serial.begin(115200);
  Wire.begin();

  // Initialize OLED display
  if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
    Serial.println("OLED initialization failed!");
    while (true); // Halt program if OLED is not found
  }

  display.clearDisplay();
  display.display();

  // Configure pin modes
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(BTN_PIN, INPUT_PULLUP); // Button connected to GND and input pin
```

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```
// Initial display message
updateOLED("System Ready");
Serial.println("System Ready");
}

// ===== MAIN LOOP =====
void loop() {
  int buttonState = digitalRead(BTN_PIN);

  // --- Button Press Detected (LOW due to INPUT_PULLUP) ---
  if (buttonState == LOW && !buttonPressed) {
    buttonPressed = true;
    pressStart = millis(); // Record the time of press
    updateOLED("Button Pressed...");
  }

  // --- Button Released ---
  if (buttonState == HIGH && buttonPressed) {
    buttonPressed = false;
    unsigned long pressDuration = millis() - pressStart; // Measure press duration

    // Decide action based on press time
    if (pressDuration > LONG_PRESS_TIME) {
      handleLongPress(); // Long press detected
    } else {
      handleShortPress(); // Short press detected
    }
  }

  delay(20); // Simple debounce delay to avoid false triggers
}
```

Explanation:

This project, **Smart Button Control with OLED Feedback**, uses a single push button (GPIO 33) to perform two different actions based on the press duration.

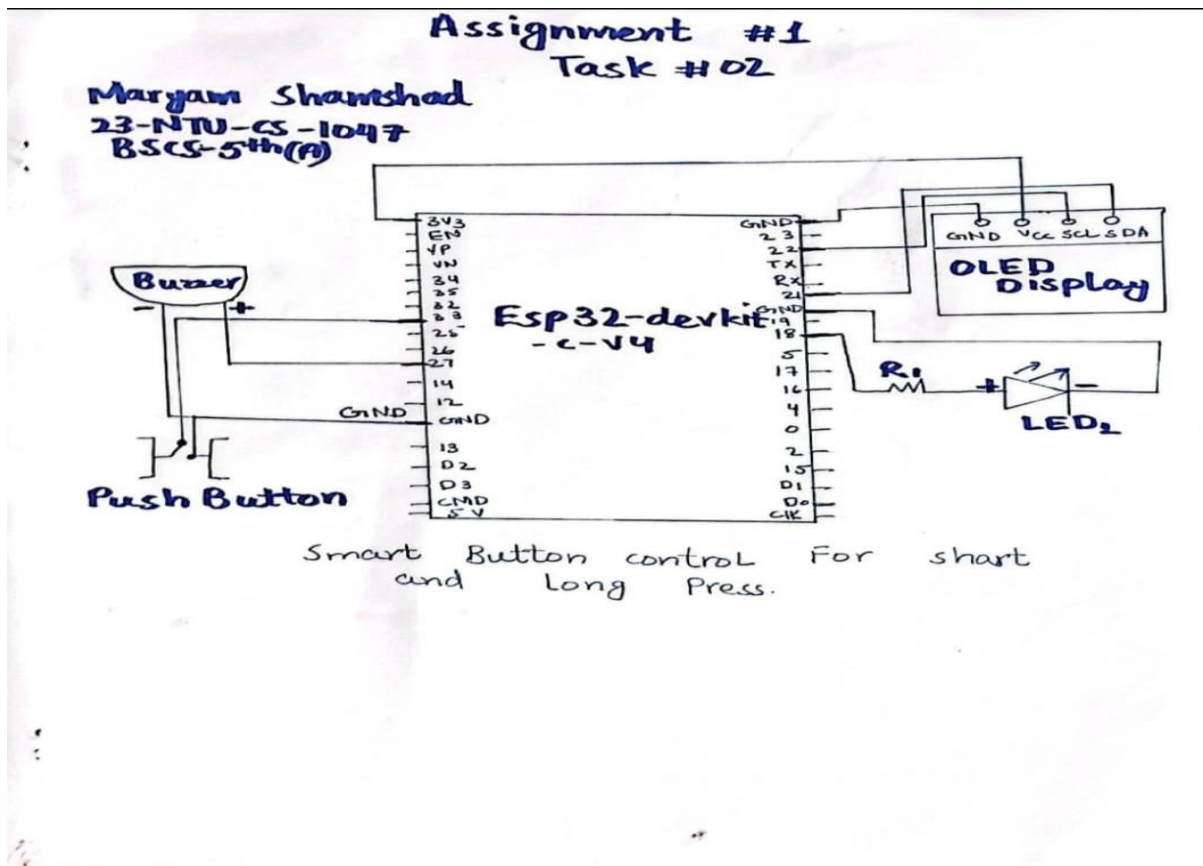
- Short press → toggle LED
- Long press (> 1.5 s) → play a buzzer tone

The buzzer (GPIO 27) produced a brief tone. **OLED display** provides real-time visual feedback by showing:

- LED is turned on or off
- buzzer is playing

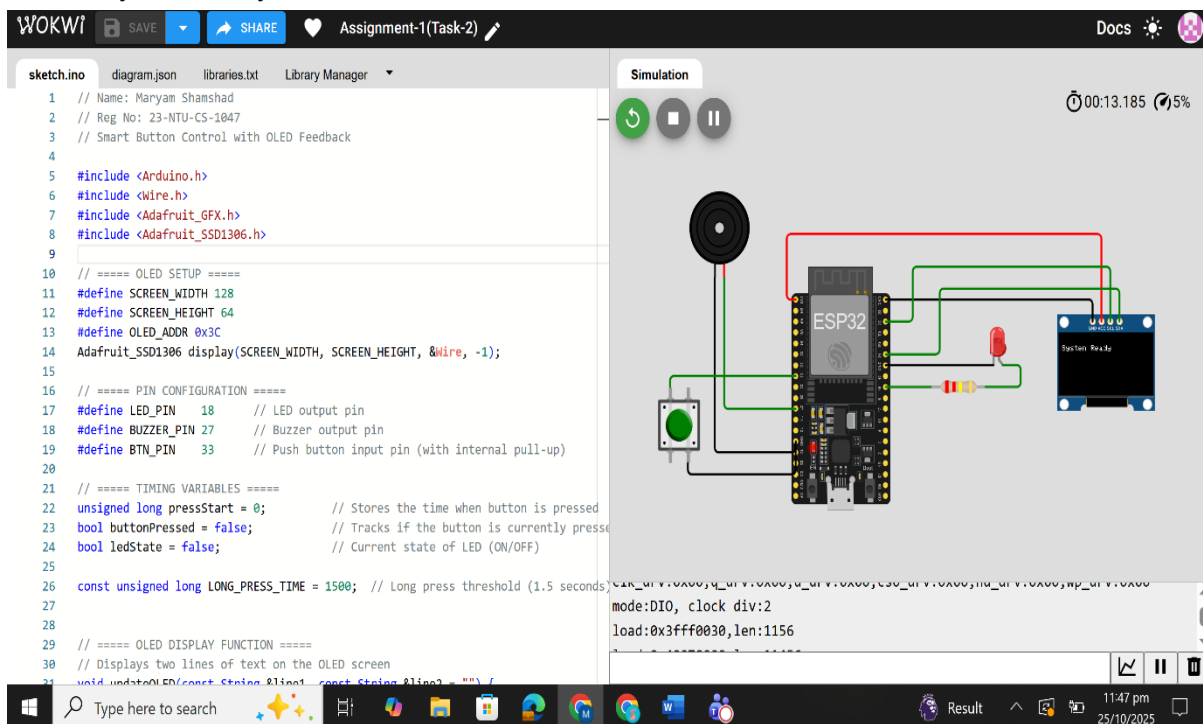
It helps us to control multiple functions with just one button, while the combination of sound and display at same time.

Sketching:



Output:

- System Ready



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- LED On

WOKWI SAVE SHARE Assignment-1(Task-2) Docs

sketch.ino diagram.json libraries.txt Library Manager

```
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29 // Displays two lines of text on the OLED screen
30 void updateOLED(const String &line1, const String &line2, "") {
31   display.clearDisplay();
32   display.setCursor(0, 0);
33   display.println(line1);
34   display.println(line2);
35   display.display();
36 }
```

Simulation

00:21.668 3%

entry 0x4008050c
System Ready
Short Press -> Toggle LED

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- LED Off

WOKWI SAVE SHARE Assignment-1(Task-2) Docs

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Simulation

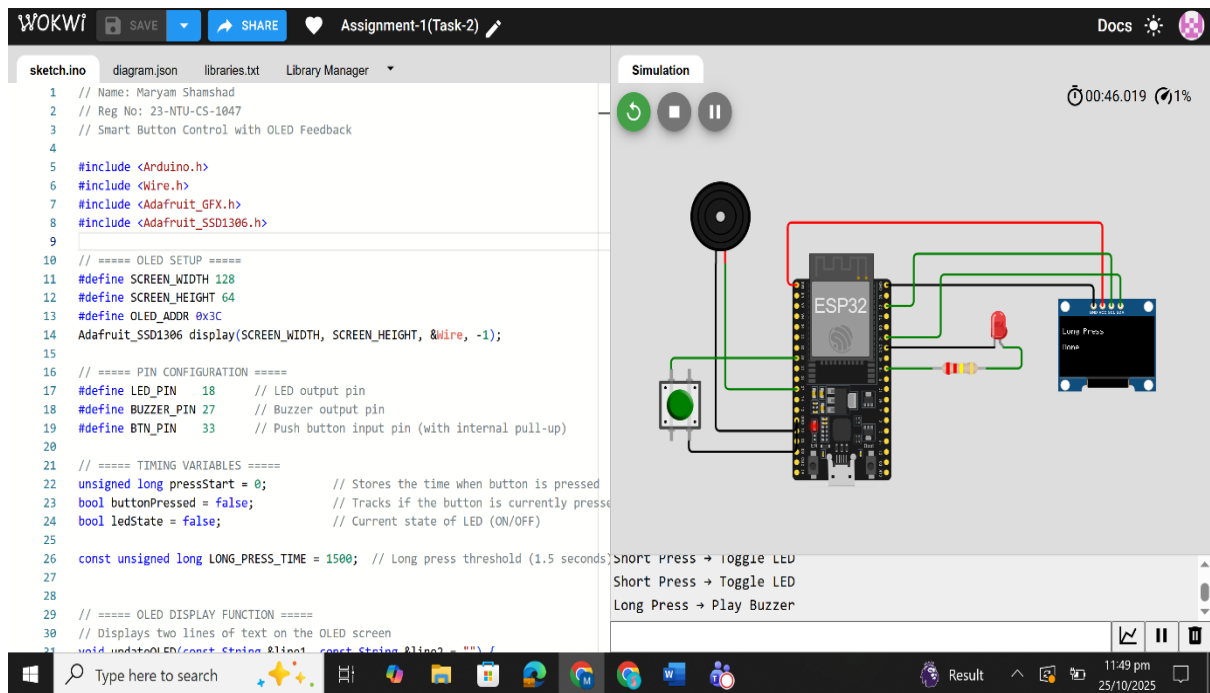
00:33.935 0%

System Ready
Short Press -> Toggle LED
Short Press -> Toggle LED

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- 4-Buzzer Tone:



Wokwi Link:

<https://wokwi.com/projects/445784915551715329>

Loom Vedio:

<https://www.loom.com/share/4cbb14e1ae8347959e66b994a8195287>